

Use Case Name:	Administer System
Scenario:	The system administrator accesses the platform to configure settings, manage user access, and monitor the overall health of the interoperability network.
Triggering Event:	The WAH4PCE Admin accesses the administrative portal.
Brief Description:	This use case outlines how an administrator interacts with the platform to ensure system uptime, configure integration endpoints, and manage administrative settings.
Actors:	Primary: WAH4PCE Admin
Related Use Cases:	None
Stakeholders:	WAH4PCE Admin: Requires comprehensive and secure access to system controls. All Users: Rely on the admin to maintain system availability and security.
Preconditions:	The WAH4PCE Admin must be authenticated with valid administrative privileges.
Postconditions:	Any system configurations made must be saved, applied, and recorded in the system audit trail.
Flow of Activities:	Actor: Admin logs into the dashboard and selects a maintenance or configuration module. System: Authenticates the Admin and retrieves the requested administrative interface. Actor: Admin inputs configuration changes or acknowledges system alerts. System: Commits the changes, updates the system state, and logs the administrative activity.

Exception Conditions:	E1: Invalid Credentials: If the Admin provides incorrect login details, the System denies access and logs the failed attempt.
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Use Case Name:	Request Health Data
Scenario:	An authorized provider or DOH system searches the interoperable repository to retrieve specific patient records.
Triggering Event:	The DOH iHomis or WAH Provider submits a search query or retrieval request via the platform.
Brief Description:	This use case governs the secure querying and extraction of interoperable health data by authorized external entities.
Actors:	Primary: DOH iHomis, WAH Provider
Related Use Cases:	None
Stakeholders:	DOH iHomis / WAH Provider: Needs timely and accurate access to patient records for care or analysis.
Preconditions:	The requesting actor must be authenticated and have the proper authorization scope to view the requested data type.
Postconditions:	The requested data is securely delivered to the actor, and the access event is recorded in the system audit logs.
Flow of Activities:	Actor: Submits a data request with specific search parameters. System: Verifies the actor's identity and checks their access authorization scope. System: Queries the repository, compiles the matching records, and encrypts the response. System: Transmits the requested data back to the Actor.
Exception Conditions:	E1: Data Not Found: If no records match the criteria, the System returns a standard '404 Not Found' or empty result set. E2: Unauthorized Scope: If the Actor requests data they are not permitted to see, the System denies the request and logs a security alert.

Use Case Name:	Manage Health Data
Scenario:	An authorized data owner updates, archives, or corrects existing health records they previously submitted to the platform.
Triggering Event:	The DOH iHomis or WAH Provider sends a modification or update command for a specific existing record.
Brief Description:	Allows primary actors to maintain the accuracy and lifecycle of their submitted data, ensuring the interoperable ecosystem contains the most current clinical information.
Actors:	Primary: DOH iHomis, WAH Provider
Related Use Cases:	None
Stakeholders:	DOH iHomis / WAH Provider: Requires the ability to amend or update patient information as clinical situations evolve.
Preconditions:	The target record must exist in the repository, and the actor must be identified as the authoritative owner of that record.
Postconditions:	The record is successfully updated, with version history maintained, and an audit trail of the change is logged.
Flow of Activities:	Actor: Submits an update payload for an existing health record. System: Validates the Actor's authority to modify this specific data asset. System: Applies the updates, creates a new version of the record, and archives the previous state. System: Sends a successful modification confirmation back to the Actor.
Exception Conditions:	E1: Record Locked/Conflict: If another process is currently accessing or modifying the record, the System rejects the update to prevent corruption and prompts the Actor to retry later.

Use Case Name:	Provide Health Data
Scenario:	A health data provider submits patient records into the WAH4PCE Interoperability platform for storage and sharing.
Triggering Event:	The DOH iHomis or WAH Provider initiates a data payload transfer to the platform's integration endpoint.
Brief Description:	This use case manages the core data ingestion process, allowing external systems to supply health records to the ecosystem. It conditionally triggers translation services if the data is not in a standard format.
Actors:	Primary: DOH iHomis, WAH Provider Secondary: WAH4PCE AI
Related Use Cases:	Extended by: Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Stakeholders:	DOH iHomis / WAH Provider: Needs a reliable and secure endpoint to submit data. Patients: Rely on the system to securely handle their submitted records.
Preconditions:	The submitting actor must be authenticated, and the WAH4PCE interoperability endpoints must be online.
Postconditions:	The health data payload is successfully ingested, processed, and a confirmation receipt is sent to the submitter.
Flow of Activities:	Actor: Submits a health data payload via API. System: Receives, encrypts, and validates the structure of the incoming payload. System: Evaluates the data format. If non-standard, it hits an extension point. System: Commits the processed data to the repository and sends a success status back to the Actor.

Exception Conditions:	E1: Malformed Payload: If the System detects an unreadable or corrupt payload, it drops the transfer and returns a 400 Bad Request error to the Actor.
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Use Case Name:	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Scenario:	An incoming data payload is detected as non-standard, requiring the AI agent to map and translate it into a standardized HL7 FHIR R4 or HL7 v2 format.
Triggering Event:	The System hits an extension point during the <i>Provide Health Data</i> process upon detecting a non-standard data schema.
Brief Description:	This extension handles the structural normalization of incoming health records via AI to ensure interoperability, while mandating a privacy consent check before completion.
Actors:	Primary: WAH4PCE AI
Related Use Cases:	WAH4PCE AI: Requires clear parameters to accurately map diverse data schemas. Requestors: Depend on this process for uniform, standardized data consumption.
Stakeholders:	WAH4PCE AI: Requires clear prompt instructions and valid API tokens. Provider: Relies on the AI to prevent them from having to manually re-type data.
Preconditions:	A non-standard health data payload must be staged in temporary memory, and the AI service must be operational.
Postconditions:	The data payload is successfully standardized into PH Core HL7 FHIR R4 or HL7 v2 without losing clinical context.
Flow of Activities:	Actor (AI): System: Routes the non-standard data payload to the AI agent.

	Actor: Analyzes the schema and performs semantic mapping to the target standard. System: Triggers the <i>Validate Privacy Consent</i> use case as a mandatory inclusion. Actor: Finalizes the standardized output and returns it to the main processing pipeline.
Exception Conditions:	E1: Translation Failure: If the AI encounters ambiguous data it cannot safely translate, it halts processing, quarantines the payload, and notifies the administrator.

Use Case Name:	Validate Privacy Consent
Scenario:	The system checks a patient's active consent directives to ensure their data is legally permitted to be processed and shared within the network.
Triggering Event:	Automatically triggered by the <i>Translate Data to/from PH Core HL7 FHIR R4/HL7 v2</i> use case.
Brief Description:	A strict, automated compliance check that cross-references patient identifiers against consent registries to approve or block the processing of health information.
Actors:	Primary: System (Internal)
Related Use Cases:	Included by: Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Stakeholders:	Patient: Requires strict adherence to their privacy and sharing preferences. WAH4PCE Admin: Needs a reliable audit trail of all consent checks for compliance.
Preconditions:	Patient identifiers must be parsable from the incoming payload, and the consent registry must be accessible.
Postconditions:	A "pass/fail" consent status is appended to the transaction, and the verification event is logged in the audit trail.
Flow of Activities:	System: Extracts patient identifiers from the active data payload. System: Queries the consent registry to retrieve the patient's current sharing permissions. System: Evaluates the permissions against the current transaction. System: Returns an authorization approval or denial back to the translating process.
Exception Conditions:	E1: Consent Revoked/Opt-Out: If the registry indicates the patient has opted out, the System immediately purges the payload, halts processing, and logs a compliance rejection.